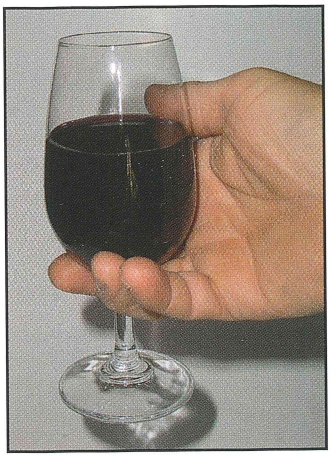
Set 27: Empirical, molecular and structural formula

Example

A barrel of wine that had been stored for some years was tested for quality before being bottled. It was found to be highly acidic and unpleasant to drink. Chemical analysis revealed that the compound responsible for the unpleasant taste contained only carbon, hydrogen and oxygen. A small amount of the compound was extracted and purified. When a 0.344 g sample of the compound was burnt in pure oxygen it produced 0.504 g of carbon dioxide and 0.206 g of water.


Wine

(a) Determine the empirical formula of the compound.

(b) A 1.876 g sample of the compound was vaporised and found to occupy 0.973 L at 120 °C and a pressure of 105 kPa.
  (i) Calculate the molecular mass of the compound.
  (ii) Determine the compound's molecular formula.

(c) Write a possible structural formula for the compound.

(a) $n(C) = n(CO_2) = \frac{m}{M} = \frac{0.504}{44.01} = 0.01145 \text{ mol}$ $\quad M(CO_2) = 44.01 \text{ g mol}^{-1}$

$m(C) = nM = 0.01145 \times 12.01 = 0.1375 \text{ g}$

$n(H) = 2n(H_2O) = 2 \times \frac{m}{M} = 2 \times \frac{0.206}{18.016} = 0.02287 \text{ mol}$ $\quad M(H_2O) = 18.016 \text{ g mol}^{-1}$

$m(H) = nM = 0.02287 \times 1.008 = 0.02305 \text{ g}$

$m(O) = m(\text{Compound}) - (m(C) + m(H))$

$= 0.344 - (0.1375 + 0.02305) = 0.1835 \text{ g}$

$n(O) = \frac{m}{M} = \frac{0.1835}{16.00} = 0.01147 \text{ mol}$

Elements	C	H	O
Number of moles	0.01145	0.02287	0.01147
Simplest ratio (Divide by smallest)	$\frac{0.01145}{0.01145}$	$\frac{0.02287}{0.01145}$	$\frac{0.01147}{0.01145}$
	1.00	1.997	1.002
Simplest whole number ratio	1	2	1

The empirical formula is therefore CH_2O

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Notes

(b)(i) Calculate the volume of the gas at STP.

$V_1 = 0.973$ L $V_2 = ?$

$P_1 = 105$ kPa $P_2 = 101.3$ kPa

$T_1 = 120$ °C $= 393$ K $T_2 = 0$ °C $= 273$ K

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2} \text{ so } V_2 = V_{STP} = \frac{P_1V_1T_2}{T_1P_2} = \frac{105 \times 0.973 \times 273}{393 \times 101.3} = 0.7006 \text{ L}$$

Calculate the number of moles of the gas.

$$n(\text{compound}) = n(\text{gas}) = \frac{V_{STP}}{22.41} = \frac{0.7006}{22.41} = 0.03126 \text{ mol}$$

OR Use $PV = nRT$ that is:

$$n(\text{compound}) = n(\text{gas}) = \frac{PV}{RT} = \frac{105 \times 0.973}{8.315 \times 393} = 0.03126 \text{ mol}$$

Calculate the molar mass.

$$n = \frac{m}{M} \text{ so } M(\text{compound}) = \frac{m}{n} = \frac{1.876}{0.03126} = 60.013 \text{ g mol}^{-1}$$

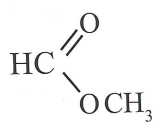
$$\text{so } M_r(\text{compound}) = 60.013$$

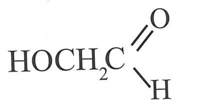
(ii) $M_r(CH_2O) = 12.01 + 2(1.008) + 16.00) = 30.026$

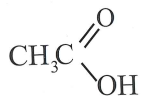
$$\frac{M_r(\text{compound})}{M_r(CH_2O)} = \frac{60.013}{30.026} = 1.9987 \approx 2$$

The molecular formula is 2 × the empirical formula = $2 \times CH_2O = C_2H_4O_2$

(c) There are a number of possible structures for a compound with this molecular formula.

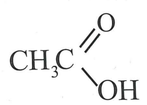






are three of them.

However, as the compound caused the wine to become acidic the compound must be an acid. So the most likely structure is:



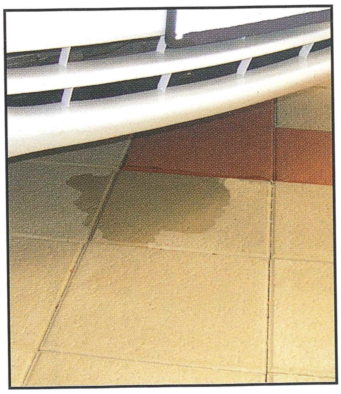
Set 27 Exercises

1. A hydrocarbon extracted from oil shale was analysed and found to contain 85.7% carbon and 14.3% hydrogen.
   (a) Determine the empirical formula of the hydrocarbon.
   (b) A 2.80 g sample of the hydrocarbon, in the gaseous state occupied 1.18 L at 25.0 °C and 105 kPa pressure. Calculate its relative molecular mass and determine the molecular formula.
   (c) Draw three possible structural formulae for the hydrocarbon.
   (d) When treated with bromine the hydrocarbon formed a 2,3-dibromohydrocarbon. Which one of the possible structures is it most likely to be.

2. Fuel from a local fuel station was reported to have been contaminated. On analysis, substantial amounts of an unidentified organic compound containing only carbon, hydrogen and oxygen was found. When 3.45 g of the purified compound was burnt in oxygen, 6.60 g of carbon dioxide and 4.05 g of water was produced.
   (a) Determine the empirical formula of the compound.
   (b) When 1.38 g of the compound was vaporised and the vapour heated to 100.0 °C it was found to occupy 0.950 L at a pressure of 98.0 kPa. Calculate its relative molecular mass and determine the molecular formula.
   (c) Draw possible structural formulae for the compound.
   (d) The compound was tested further and found to react with sodium. Draw its structural formula.

Question 2: Fuel station

3. Liquid, seen leaking from a car was analysed and was found to contain large amounts of an organic compound made up of only carbon, hydrogen and oxygen. When 0.682 g of the purified compound was burnt in oxygen, 0.968 g of carbon dioxide and 0.594 g of water were produced. Another 0.744 g of pure compound was vaporised. It was found the occupy 497 mL at a temperature of 200.0 °C and a pressure of 95.0 kPa.
   The compound was found to react with sodium to produce hydrogen but analysis of its infrared spectrum indicated that the compound did not contain a carbonyl group.
   (a) Determine the empirical formula of the compound.
   (b) Calculate its relative molecular mass and determine the molecular formula.
   (c) Draw two possible structural formulae for the compound.



Question 3: Fluid leak

4. During the production of propan-2-ol a strong smelling by product was detected in the reaction mixture. Contamination of the reaction mixture was suspected. To track down the contaminant the unknown compound was isolated, purified and when analysed found to contain carbon, hydrogen, nitrogen and possibly oxygen.
   When 1.180 g sample of the compound was burnt in oxygen it produced 1.758 g of carbon dioxide and 0.900 g of water. A second 1.180 g sample was decomposed to release 0.471 L of nitrogen gas measured at a temperature of 25.0 °C and a pressure of 105 kPa.
   Another sample of mass 0.5896 g was vaporised and was found to have a volume of 0.281 L at a temperature of 50.0 °C and a pressure of 95.5 kPa.
   (a) Determine the empirical formula of the compound.
   (b) Calculate its relative molecular mass and determine the molecular formula.
   (c) Draw a structural formula for the compound.

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Question 5: Toilet deodorant

Notes

5. Analysis of a toilet deodorant found that it contained a large proportion of a compound that contained only carbon, hydrogen and chlorine. 1.324 g of the compound was divided into two equal samples. The first sample, when burnt in oxygen produced 1.189 g of carbon dioxide. The second sample was oxidised with concentrated nitric acid and treated with silver nitrate solution to produce 1.292 g of silver chloride. The relative molecular mass of the unknown compound was found from mass spectrometer measurements to be 147.
   (a) Determine the empirical formula of the compound.
   (b) Determine the molecular formula.
   (c) Nuclear magnetic resonance analysis indicated the presence of a benzene structure. Draw three possible structural formulae for the compound.

6. During the routine analysis of coal tar, a by product of the production of coke from coal, a colourless, oily liquid was isolated. Analysis revealed that it contained only carbon, hydrogen and nitrogen. When a 0.620 g sample of the compound was burnt in oxygen it produced 1.76 g of carbon dioxide. A second sample of the compound of mass 0.232 g was decomposed. This produced 29.5 mL of nitrogen gas measured at 15.0 °C and 101.3 kPa pressure. Another sample of the compound was vaporised and at 100 °C and 101.3 kPa pressure was found to have a density of 3.04 g L^{-1}. The unknown compound was found to undergo substitution reactions with bromine and exhibit basic properties in aqueous solution.
   (a) Determine the empirical formula of the compound.
   (b) Determine the molecular formula.
   (c) Draw the structural formula for the compound.

7. An environmental chemist was asked to determine the identity of the solvent used in an insecticide preparation. Preliminary analysis revealed that the compound contained only carbon, hydrogen and oxygen. A 0.666 g sample of the compound when burnt in oxygen produced 1.584 g of carbon dioxide and 0.810 g of water. The unknown compound was vaporised and at a temperature of 200 °C and a pressure of 101.3 kPa its density was measured to be 1.91 g L^{-1}.
   (a) Determine the empirical formula of the compound.
   (b) Determine the molecular formula.
   (c) Draw three possible structural formulae for the compound.
   (d) The compound reacts slowly with sodium, but it is not oxidised by acidified potassium permanganate solution. Draw the structural formula for the compound.

8. An organic compound used in the manufacture of complex experimental pesticides contains carbon, hydrogen, nitrogen and sulfur only.
   A 3.127 g sample of the compound was burnt completely in excess pure oxygen. At 100 °C all the products were gases and had a total mass of 12.56 g. Analysis of the gaseous mixture indicated that there were 5 gases present. They were identified as water, carbon dioxide, nitrogen dioxide, sulfur dioxide and oxygen.
   The gas mixture was passed through silica gel which absorbs water. The mass of the remaining gases was found to be 11.518 g. These remaining gases were then cooled in an ice bath. The nitrogen dioxide condensed to a liquid and was removed. The mass of the remaining gases was 10.454 g.

Notes

The sulfur dioxide was converted to sulfate ion by shaking the remaining gas mixture with chlorine water. Excess barium chloride solution was then added. The resulting precipitate of barium sulfate was filtered, dried and found to have a mass of 5.398 g.

(a) Determine the empirical formula.

(b) Another 2.372 g sample of the compound was vaporised and found to have a volume of 415.5 mL at 103.6 kPa and 22.0 °C. Calculate the molecular mass and determine the molecular formula of the compound.

(c) Further tests indicate that this is an aromatic compound containing a benzene ring. Draw a possible structure for this compound.

9. 0.682 g of a compound containing carbon, hydrogen and oxygen produced 0.968 g of carbon dioxide and 0.594 g of water when burnt in oxygen. A further 0.744 g sample of the compound, when vaporised, occupied 497 mL at 200.0 °C and 95 kPa pressure. The unknown compound reacted with sodium to produce hydrogen, but analysis of its infrared spectrum indicated that the compound did not contain a carbonyl group. Determine:

(a) the empirical formula

(b) the molar mass and the molecular formula

(c) two possible structural formulae

(d) The compound reacts slowly with sodium, but it is not oxidised by acidic potassium permanganate. Write the structural formula for the compound.

10. A 0.666 g sample of an unknown organic compound when burnt in oxygen produced 1.584 g of carbon dioxide and 0.810 g of water. The unknown, in the gaseous state, had a density of 1.91 g L^{-1} at 200.0 °C and 101.3 kPa pressure. Determine:

(a) the empirical formula

(b) the molecular formula

(c) three possible structural formulae

(d) The compound reacts slowly with sodium, but it is not oxidised by acidic potassium permanganate. Write the structural formula for the compound

11. An organic compound used in perfumes and flavours contains only carbon, hydrogen and oxygen. When 1.760 g of the compound was burnt in oxygen, it produced 3.52 g of carbon dioxide and 1.44 g of water. When vaporised, the compound has a density of 2.62 g L^{-1} at 150.0 °C and 105.0 kPa pressure. Determine:

(a) the empirical formula

(b) the molecular formula

(c) the structural formula if the compound is an ester

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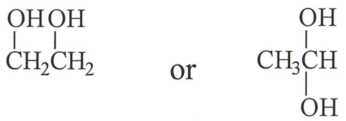
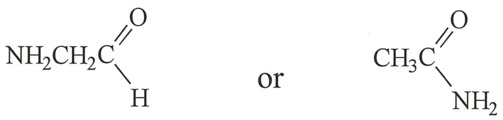
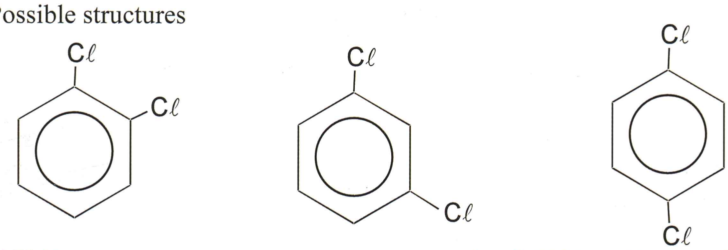
1. (a) CH_2 (b) C_4H_8
   (c) $CH_3CH_2CH{=}CH_2$ $CH_3CH{=}CHCH_3$ $CH_3C(CH_3){=}CH_2$
   (d) Bromine adds to atoms either end of the double bond so the double bond is between atoms 2 and 3 therefore formula is $CH_3CH{=}CHCH_3$ that is but-2-ene
2. (a) C_2H_6O
   (b) $M_r(C_2H_6O) = 46.068$ Relative molecular mass is the same as the relative empirical formula mass so the molecular formula is C_2H_6O
   (c) Possible structures include: CH_3CH_2OH and CH_3-O-CH_3
   (d) As the compound reacts with sodium it is most likely an alcohol that is CH_3CH_2OH
3. (a) CH_3O (b) $M_r = 61.975$, C_2H_6O
   (c) As the compound reacts with sodium it is most likely an alcohol so possible structures are.



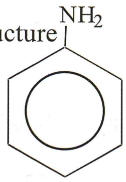
4. (a) C_2H_5NO
   (b) $M_r(C_2H_5NO) = 59.007$ Molecular formula C_2H_5NO
   (c) Possible structures for a compound with this molecular formula include.



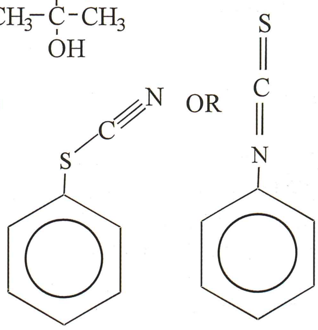
5. (a) $C_3H_2C\ell$ (b) $M_r(C_3H_2C\ell) = 73.496$, $C_6H_4C\ell_2$
   (c) Possible structures



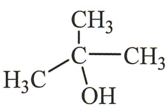
6. (a) C_6H_7N (b) $M_r(C_6H_7N) = 93.126$, C_6H_7N (c) A possible structure



7. (a) $C_4H_{10}O$ (b) $M_r(C_4H_{10}O) = 74.12$, $C_4H_{10}O$
   (c) Possible structures include
   $CH_3CH_2CH_2CH_2OH$ $CH_3CH_2CH(OH)CH_3$ $CH_3C(CH_3)(OH)CH_3$ $CH_3CH(CH_3)CH_2OH$
   (d) Based on the observed reactivity's the structure of the compound is $CH_3C(CH_3)(OH)CH_3$
8. (a) C_7H_5NO (b) $M_r(C_7H_5NS) = 135.18$, C_7H_5NS (c) OR



9. (a) CH_3O (b) MFM = 62.0 g mol^{-1}, $C_2H_6O_2$
10. (a) $C_4H_{10}O$ (b) $C_4H_{10}O$
    (c) $CH_3CCH_3OHCH_3$, $CH_3CH_2CH_2CH_2OH$, $CH_3CHOHCH_2CH_3$,
    (d)



11. (a) C_2H_4O (b) $C_4H_8O_2$
    (c) $CH_3COOCH_2CH_3$ or $CHCOOCH_2CH_2CH_3$ or $CH_3CH_2COOCH_3$